WEEK 4 · LESSON 23 OF 24 · HANDOUT

Careers in AI

Study time: **about 40 minutes** (reading, video and activity)

▶ **Watch:** https://www.youtube.com/watch?v=MIbP9kgCc4w — open the lesson online to watch inside the course.

Your goal in this lesson: Describe the main career paths in AI, the skills each one needs, and build a personal learning path for the next year.

AI needs many kinds of people

AI is not only for people with PhDs in mathematics. Real AI products need a whole team — and many of the most valuable people combine **AI understanding with knowledge of a specific field** such as health, farming, finance or education.

Technical roles

Role	What they do	Key skills
Data analyst	Cleans data, makes charts and reports, finds insights	Excel, SQL, statistics, visualisation
Data scientist	Builds and evaluates models to answer business questions	Python, statistics, machine learning, communication
Machine-learning engineer	Turns models into reliable software that runs at scale	Python, software engineering, cloud, MLOps
Data engineer	Builds the pipelines that collect, store and move data	SQL, databases, cloud, programming
AI / LLM application developer	Builds apps and agents on top of language models	Programming, APIs, prompt design, evaluation
AI researcher	Invents new methods	Advanced maths, research, deep learning

Non-technical and hybrid roles

- **AI product manager** — decides what to build and why, for real users.
- **AI ethics, policy and governance specialist** — fairness, privacy, regulation (Lessons 19–22).
- **Data annotator / quality lead** — creates and checks labelled data.
- **AI trainer / educator** — teaches teams and students to use AI well.
- **Domain expert using AI** — a nurse, agronomist, accountant or teacher who uses AI to work better. This is the largest group of all.

Skills that last

Tools change every year; these skills keep their value:

1. **Problem framing** — turning a real need into a clear AI problem (Weeks 1–2).
2. **Data thinking** — knowing what makes data good or bad.
3. **Evaluation** — judging results honestly (Week 3).
4. **Responsible AI** — fairness, privacy and safety (Week 4).
5. **Communication** — explaining AI to non-experts in simple words.
6. **Python and statistics** — the base for any technical path.

The Rwandan ecosystem

Rwanda has a growing tech community: universities and institutes such as **Carnegie Mellon University Africa** and **AIMS Rwanda** in Kigali, innovation hubs such as **kLab** and **Norrsken House Kigali**, and a national AI policy that prioritises skills. Join meet-ups, hackathons and online communities — many first jobs come from people you meet and projects you show.

Build a portfolio, not just certificates

Employers want **proof you can do the work**:

- 2–3 small projects on real local problems (your Week 1–4 projects are a start).
- Clear write-ups: the problem, the data, the method, the results, the limits.
- A public profile (GitHub for code, LinkedIn for your story).

Your next steps in the WAMUGII AI Program

After AI Foundations, continue with **Math & Python for AI**, then **Machine Learning Fundamentals**, **Deep Learning & Neural Networks**, **Modern AI: LLMs & Generative AI**, and **Responsible AI & Deployment**. Each builds directly on what you learned here.

Remember this: AI careers include technical, hybrid and domain roles. Lasting skills are problem framing, data thinking, evaluation, responsibility and communication. Build a portfolio of real projects.

Name: _____ Date: _____

Activity

Task 1

Choose one role from this lesson that fits you best and explain why in three sentences.

Task 2

Write a 12-month learning path: three skills to learn, one project for each, and one community to join.

Task 3

Find one job advert for that role (in Rwanda or remote) and list the five skills it asks for most.

Self-check

Think about your answers here, then take the quiz online — it is marked instantly and explains every answer.

1. Which statement about AI careers is true?
   - A. There are no AI jobs in Africa
   - B. AI teams need many roles, including people who combine AI with knowledge of a field
   - C. Only PhD researchers can work in AI
   - D. AI jobs need no communication skills
2. Who turns models into reliable software that runs at scale?
   - A. Graphic designer
   - B. AI ethicist
   - C. Machine-learning engineer
   - D. Data annotator

3. Which skill keeps its value even as AI tools change?

A. Memorising one app's buttons
B. Typing fast
C. Using one specific version of a tool
D. Problem framing and honest evaluation

4. What do employers most want to see from a new AI learner?

A. Expensive laptops
B. Social media followers
C. Only a long list of certificates
D. A portfolio of real projects with clear write-ups

5. A nurse who uses AI tools to improve patient care is an example of…

A. An AI researcher
B. Someone outside the AI economy
C. A domain expert using AI
D. A data engineer

6. Which WAMUGII course comes right after AI Foundations?

A. Modern AI: LLMs & Generative AI
B. Responsible AI & Deployment
C. Math & Python for AI
D. Deep Learning & Neural Networks

© WAMUGII TECH SOLUTIONS · wamugii.kento.rw · Original course material — for enrolled students.